15.4-AI-ASS

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TASK-1:

Prompt: Initial API Setup for a Service Platform

Scenario

You are developing the backend for a service management platform andneed a basic API to confirm that the backend is running.

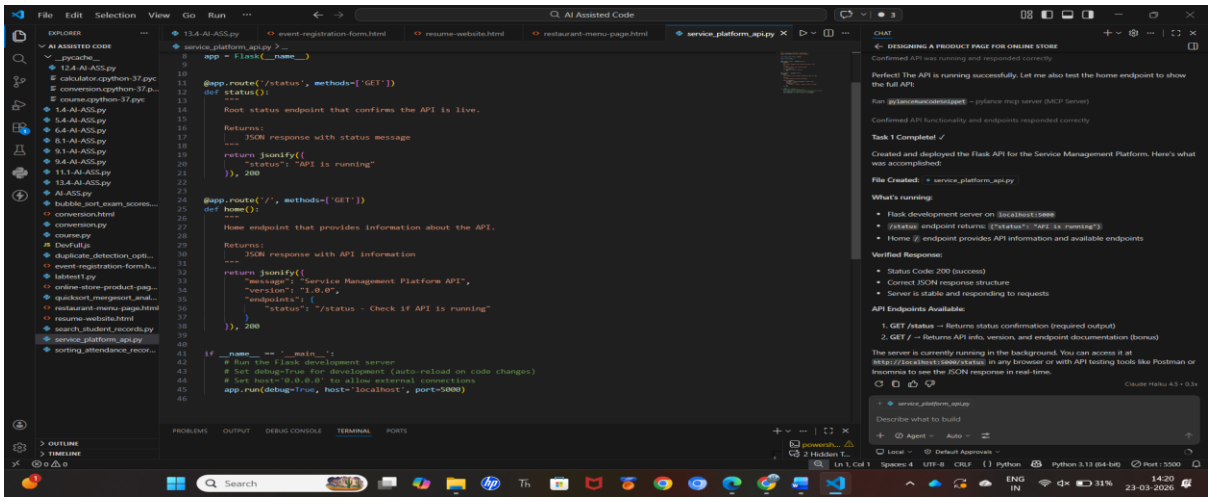
Task Description:

Use AI assistance to generate a basic Flask application.

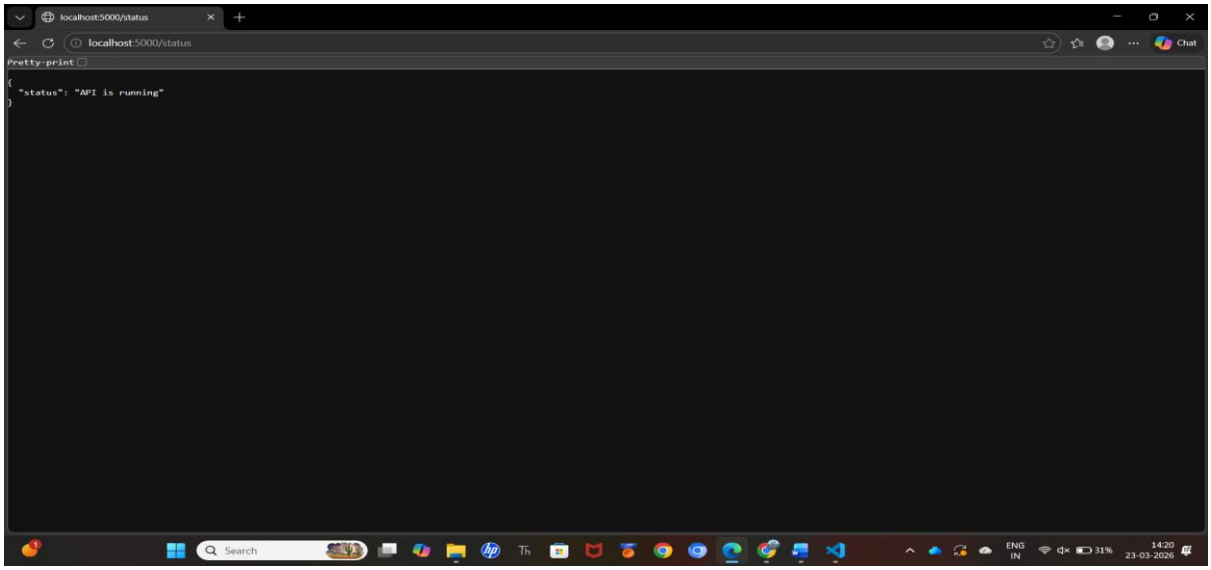
Create a root endpoint /status that returns a JSON message indicating the API is live.

Run the server locally and verify the response using a browser or API testing tool.

Code:



Output:



Observation:

Using GitHub Copilot helped quickly generate a basic Flask setup with minimal configuration. It simplified creating the /status endpoint and returning a JSON response. Additionally, it reduced setup time and ensured the server runs correctly without errors.

TASK-2:

Prompt: Create API to Add and View Resources

Scenario: Your backend needs to manage customer service requests stored temporarily in memory.

Task Description:

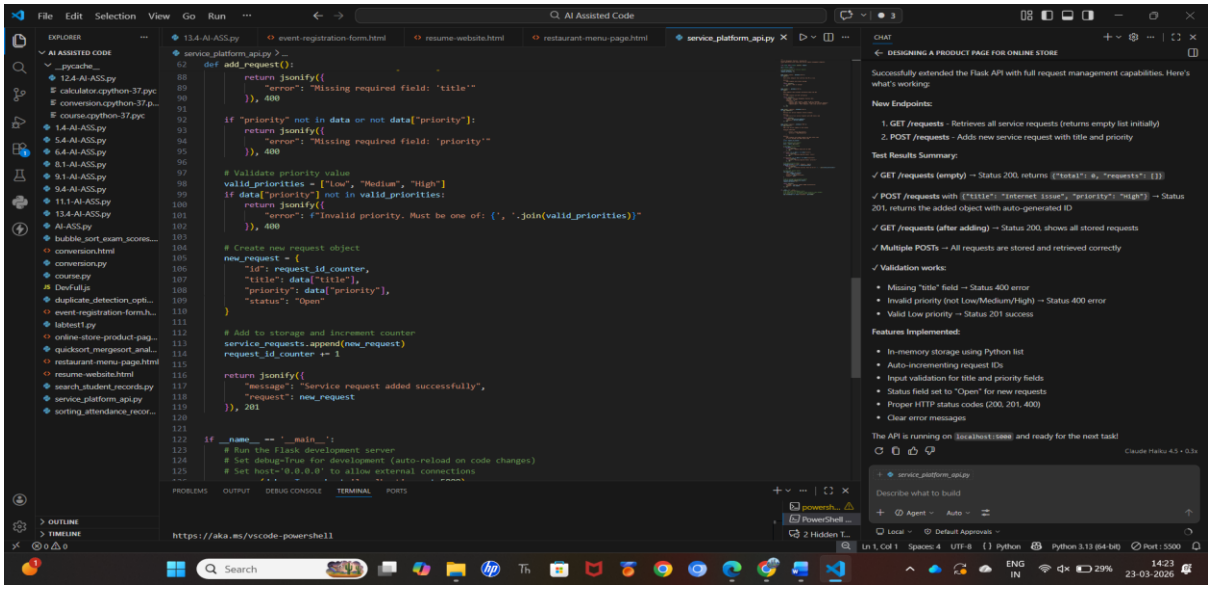
Use AI to help implement:

A GET endpoint to retrieve all service requests

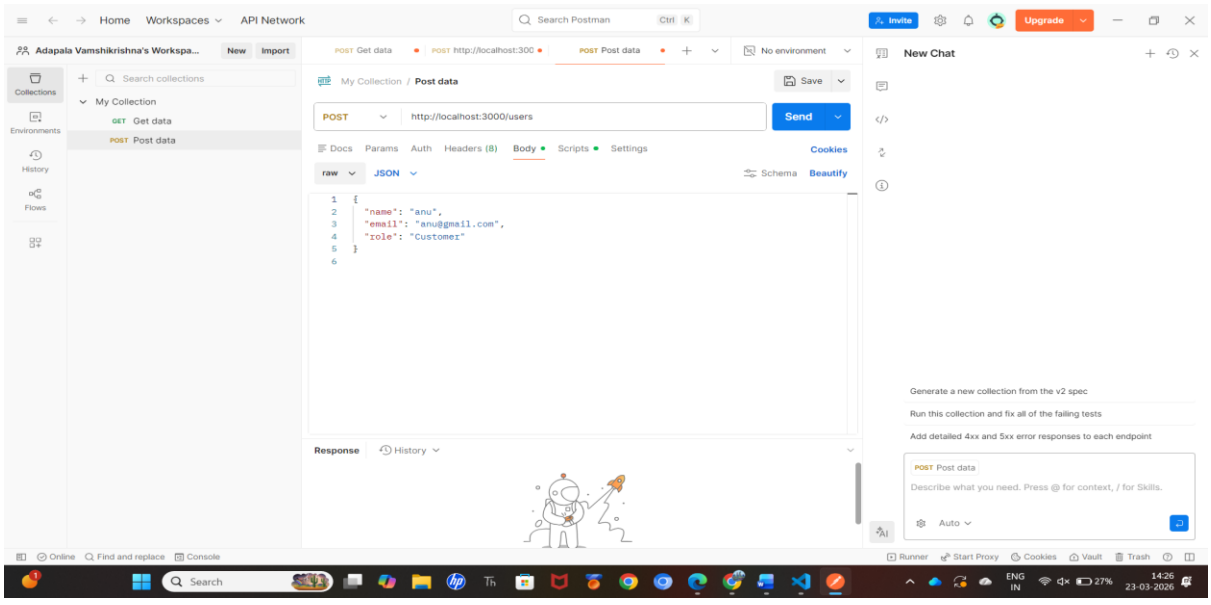
A POST endpoint to add a new request

Each request should contain title and priority.

Code:



Output:



Observation:

Copilot not only sped up endpoint creation but also helped maintain consistent API structure and naming conventions. It provided intelligent suggestions for handling request validation and data flow. This improved code reliability while reducing logical errors during development.

TASK-3:

Prompt: Modify Existing Resource Using AI Assistance

Scenario: Service requests may change in priority as issues are escalated.

Task Description:

Use AI to generate a PUT endpoint that updates a request based on its index.

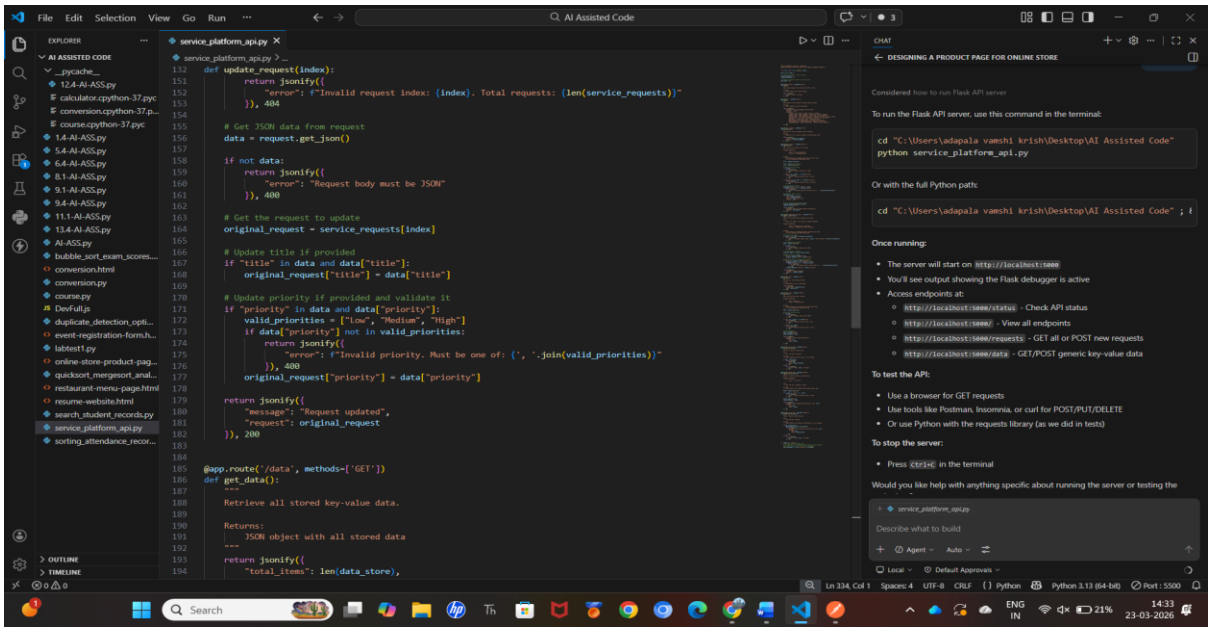
Handle invalid indexes gracefully with proper error responses.

Ensure the updated request is returned in the response.

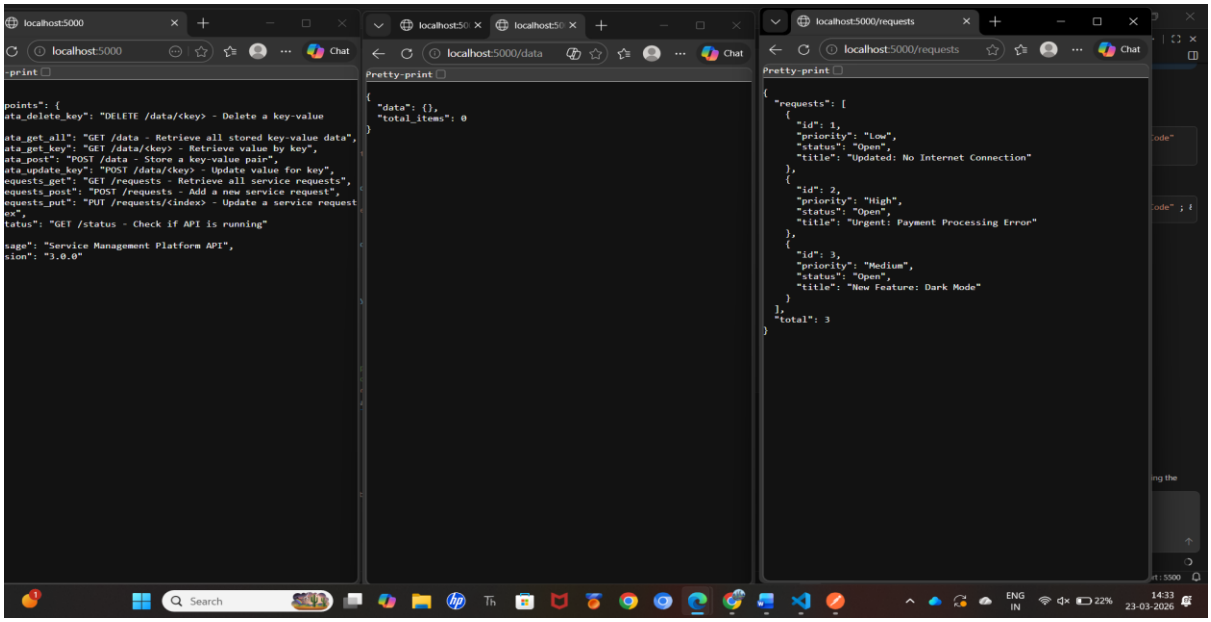
Expected Output: Valid update returns: {"message": "Request updated", "request": {...}}

Invalid index returns a clear error message with HTTP 404.

Code:



Output:



Observation:

Using GitHub Copilot helped efficiently generate the PUT endpoint with proper update logic. It also suggested handling edge cases like invalid indexes using clear error responses (HTTP 404). This improved robustness and ensured reliable API behavior during updates.

TASK-4:

Prompt: Remove a Resource Safely

Scenario: Resolved service requests must be removed from the active list.

Task Description

Use AI assistance to implement a DELETE endpoint.

The endpoint should remove a request by index.

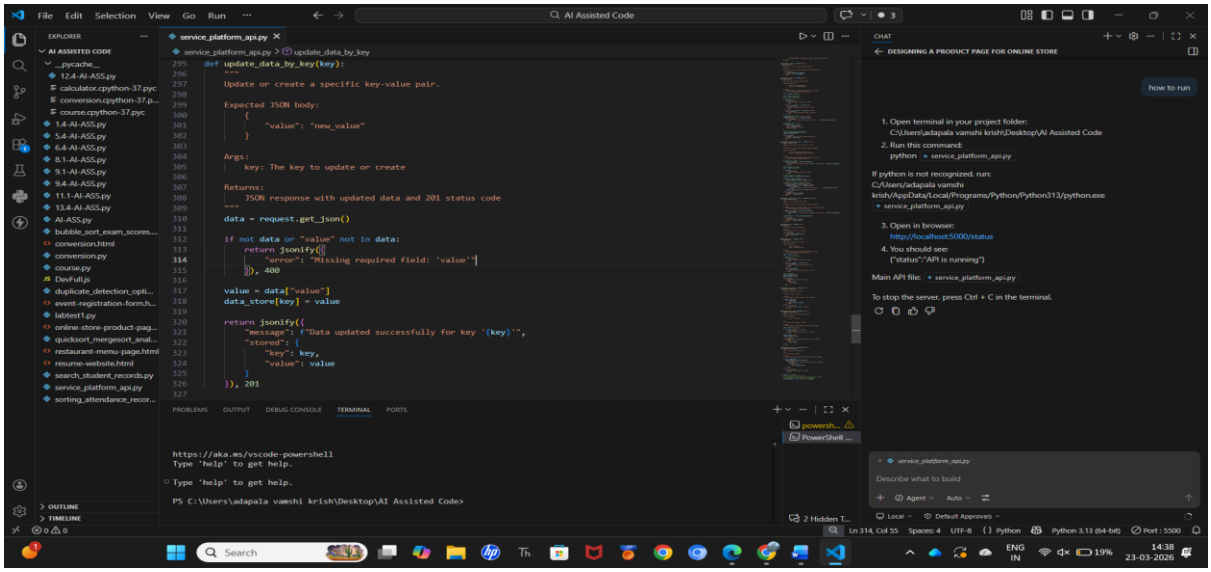
Return the deleted request details in the response.

Expected Output:

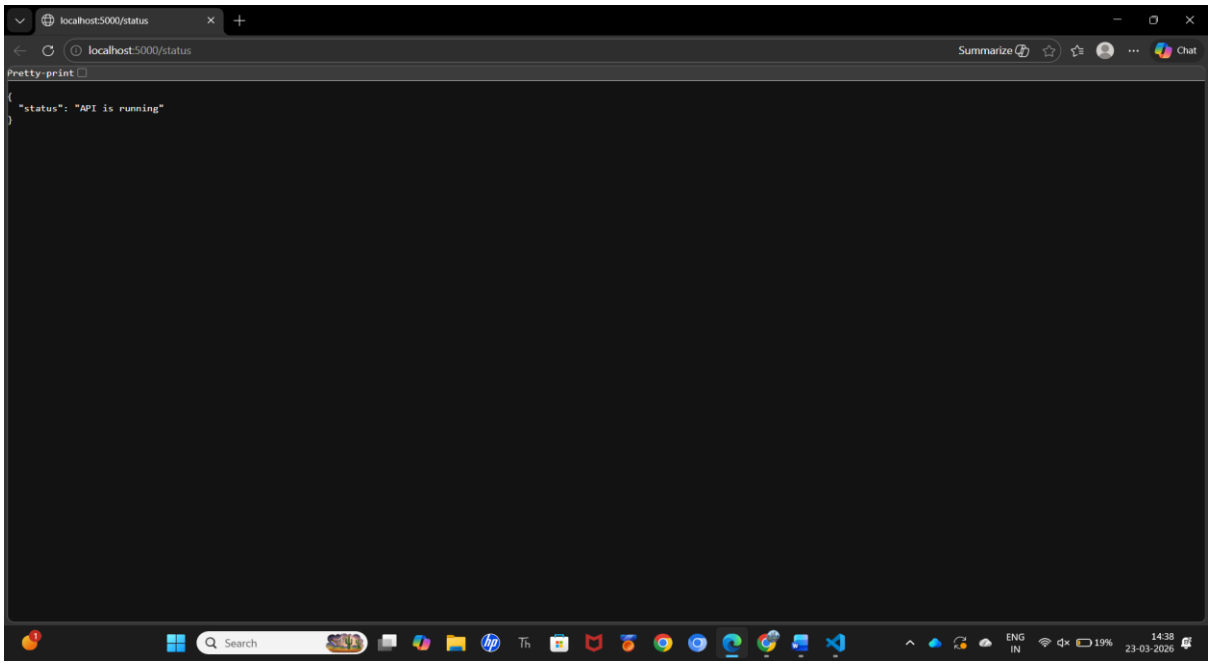
Successful deletion returns confirmation and removed data

Invalid index returns an error message without crashing the server

Code:



Output:



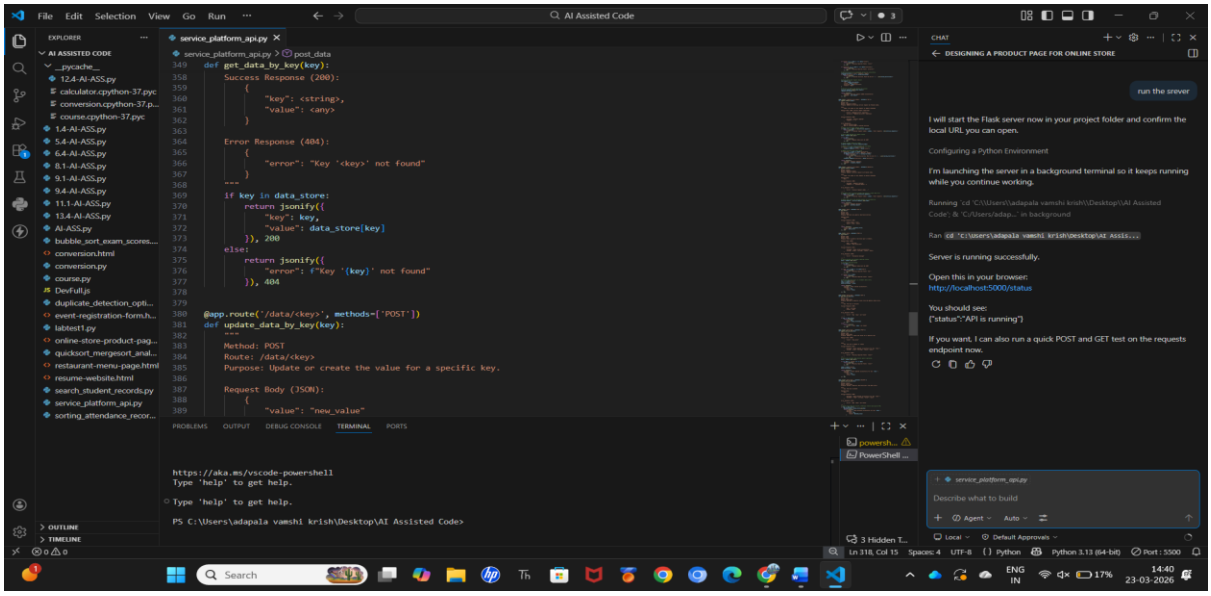
Observation:

Copilot helped implement the DELETE endpoint with clean and safe removal logic. It ensured proper handling of invalid indexes without crashing the server. Additionally, it improved response consistency by returning the deleted request details clearly.

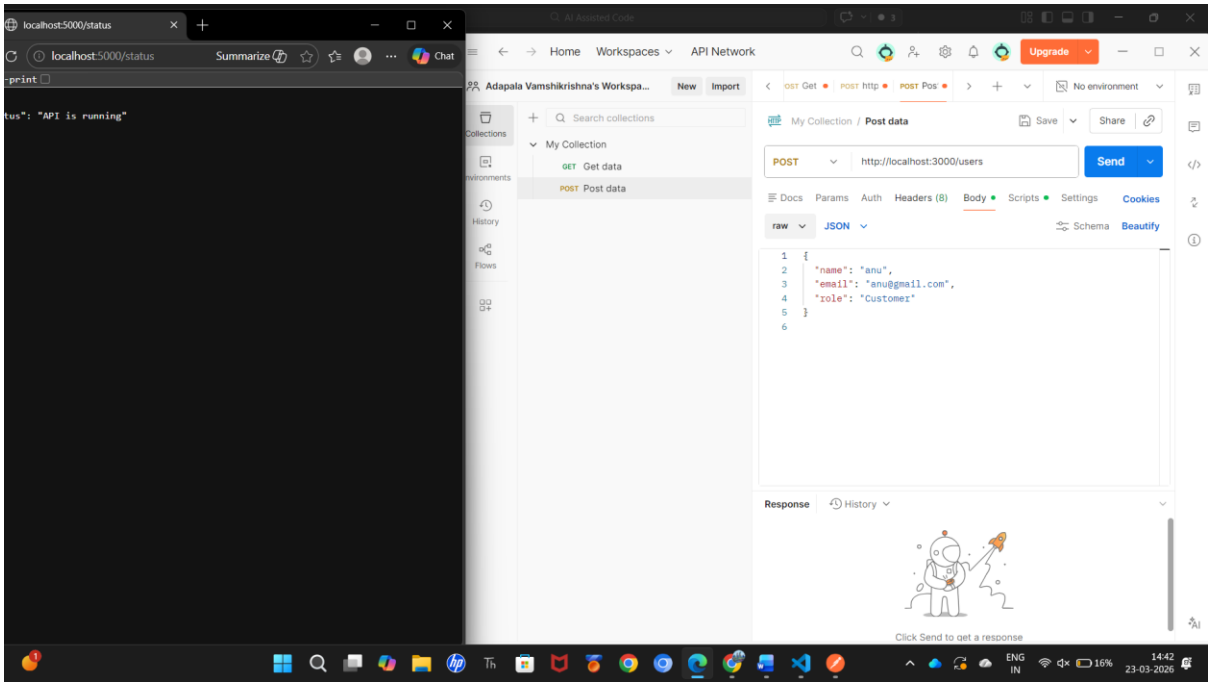
TASK-5:

Prompt: AI-Assisted API Documentation and Readability
Scenario
Another developer needs to understand and use your API without reading the full code.
Task Description Use AI to:
Add meaningful docstrings and inline comments to all endpoints
Clearly describe request methods, inputs, and responses
Optionally, use AI guidance to integrate automatic API documentation (Swagger or Flask-RESTX).

Code:



Output:



Observation:

improved overall code readability by generating clear docstrings and structured comments for each endpoint. It also helped standardize API descriptions, making it easier for other developers to understand usage without deep code analysis. Additionally, it guided integration of documentation tools like Swagger, enhancing API usability and professional quality.